Parsa Oryani

M.Sc. in Computer Engineering | Tehran, Iran

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Research Profile

I study the security and privacy of LLM-based agents that use external tools and retain information across tasks. I am interested in how untrusted inputs shape agent decisions, when sensitive information can cross security boundaries, and how to keep tool actions within the permissions a user has granted. I compare the assumptions behind proposed defenses and use that work to frame research questions about what an agent can do safely at runtime.

Education

Sharif University of Technology Sep 2025 – Present
M.Sc. in Computer Engineering, Secure Computing Tehran, Iran

- National Master's Entrance Examination (Computer Engineering, 2025): Rank 56 (estimated pool of 18,000 candidates).
- Selected coursework: Applied Cryptography (18.9/20), Deep Learning (18.35/20, Dr. Soleymani), Secure Software Systems (18.2/20), Formal Methods in Information Security, and Foundations and Applications of Blockchain.
- Deep Learning: Transformers, LLMs, state-space and generative models, self-supervised learning, multimodal foundation models, and model evaluation.

Amirkabir University of Technology (Tehran Polytechnic) Sep 2020 – Feb 2025
B.Sc. in Computer Science Tehran, Iran

- National University Entrance Examination (Mathematics, 2021): Rank 343 out of 142,062 candidates.
- Cryptography I (18.75/20) and Special Topics in Cryptography (18.25/20, highest grade in class).

Research Experience

Research Assistant, Agentic AI Security and Privacy Sep 2025 – Present
Sharif University of Technology, supervised by Dr. Morteza Amini Tehran, Iran

- Compare threat models, assumptions, and limits of recent defenses for LLM agents.
- Develop research questions on how to specify and evaluate runtime authorization for agent tool calls.
- Previously studied state verification and correctness properties in Layer-2 and cross-chain protocols.

Teaching Experience

Teaching Assistant

Computer Networks Feb – Jun 2026
Sharif University of Technology • Dr. Sadeghzadeh
Designed theory and practical networking exercises, graded coursework and quizzes, and helped students with assignments.

Machine Learning Feb – Jun 2026
Sharif University of Technology • Dr. Motahari
Prepared theory questions and coding assignments and helped grade assignments and in-class tests.

Artificial Intelligence and Lab Feb – Jun 2025
Amirkabir University of Technology • Dr. Ghatee • Undergraduate
Developed written exercises and lab tasks, reviewed student solutions, and helped grade quizzes.

Design and Analysis of Algorithms Sep 2023 – Jan 2024
Amirkabir University of Technology • Dr. Ardeshir Dolati • Undergraduate
Wrote algorithm design problems, reviewed student solutions, and helped grade coursework.

Foundations of Matrices and Linear Algebra Sep 2023 – Jan 2024
Amirkabir University of Technology • Dr. Najafi • Undergraduate
Helped students work through linear algebra problems and understand course material.

Advanced Programming Feb – Jun 2023
Amirkabir University of Technology • Dr. Bejani • Undergraduate
Led programming workshops, designed hands-on exercises, and graded code submissions.

Technical Research Toolkit

Agentic AI & LLM Systems	LLM agents, multi-agent systems, tool, MCP, and skill integration, RAG, fine-tuning, prompting, inference, agent memory, and evaluation.
Security & Formal Methods	Threat modeling, information-flow control, authorization, least privilege, provenance, taint and data-flow analysis, security invariants, runtime enforcement, SMT, model checking, and formal verification.
Deep Learning	PyTorch, Transformers, attention, LLMs, state-space models, generative models, self-supervised learning, multimodal models, and model evaluation.
Cryptography & Distributed Systems	Applied cryptography, commitments, Merkle structures, zero-knowledge proof concepts, consensus, blockchain, Layer-2, and cross-chain protocols.
Engineering	Python, Rust, Solidity, C++, PostgreSQL, Linux, Git